Optional Knowledge-Graph Layer

Optional Knowledge-Graph Layer I

An optional, **LLM-gated** stage lifts the corpus from bibliometrics to hypothesis-level evidence. For each eligible record, a local language model (Ollama, default model `gemma3:4b`) extracts structured *assertions*. Each assertion encodes a direction (supports, contradicts, or neutral), a confidence score, and a short natural-language justification against one of the 4 hypotheses declared in configuration. Assertions are serialized as RDF-compatible nanopublications (Kuhn et al. 2016) and scored by a citation-weighted evidence function.

Assertion Model I

Each assertion a encodes:

Assertion Model I

- **Direction**: supports, contradicts, or neutral with respect to a hypothesis H
- **Confidence**: a score $c_a \in [0, 1]$ from the LLM
- **Citation weight**: $\log(1 + n_{\text{cites}})$, where n_{cites} is the citation count of the asserting paper

Assertion Model I

The evidence score for hypothesis H is:

$$\text{score}(H) = \frac{\sum_{a \in A(H)^+} c_a \cdot \log(1 + n_{\text{cites}}(a)) - \sum_{a \in A(H)^-} c_a \cdot \log(1 + n_{\text{cites}}(a))}{\sum_{a \in A(H)} c_a \cdot \log(1 + n_{\text{cites}}(a))}$$

where $A(H)^+$ is the set of supporting assertions and $A(H)^-$ the contradicting ones. The score ranges from -1 (all evidence contradicts) to $+1$ (all evidence supports).

Incremental Extraction I

Assertion extraction is **incremental and resumable**: assertions are appended to `nanopublications.jsonl` at configurable checkpoint intervals (default: 50 papers). On restart, already-processed papers are skipped automatically, so a long extraction run that is interrupted can resume without re-processing. The `--clear-assertions` flag discards previous results for a fresh start.

Gating and Defaults I

This stage is optional and gated by language-model availability. With no language model configured, the hypothesis evidence scores read *pending*. When Ollama is explicitly configured and the extraction stage completes, scores are populated from the recorded, citation-weighted assertions. The hypotheses themselves, including their names and scope, come from configuration and are reported regardless of whether scoring has run.

The hypotheses explored in this instance are: H1 JWST Atmospheric Characterization; H2 Molecular Diversity Detection; H3 Cross-Method Consistency; H4 Theoretical-Observational Agreement.